Drilling Jar

DOUBLE-ACTING HYDRAULIC / MECHANICAL DRILLING JAR

JARRING

GENERAL DESCRIPTION

HYDRAULIC-DELAY JAR FOR BOTH UPSTROKE AND DOWNSTROKE, WITH A MECHANICAL LATCH, BUILT FOR COMPLEX DRILLING PROGRAMS. THE DOWNSTROKE DELAY MAKES IT EFFECTIVE IN DIRECTIONAL, HORIZONTAL AND EXTENDED-REACH WELLS. BECAUSE NO BLEED-THROUGH OPERATION IS NEEDED WHEN TAGGING OR PICKING UP OFF BOTTOM, IT CAN BE PLACED IN COMPRESSION IN THE DRILLSTRING. LATCH RELEASE SETTINGS ARE ADJUSTABLE.

TECHNICAL CHARACTERISTICS

Tool O.D. (in)	4-1/4	4-3/4	6-1/4	6-1/2	6-3/4	7	8	9-1/2
Tool O.D. (mm)	108	121	159	165	171	178	203	241
Inside Ø (in)	2	2-1/4	2-1/4	2-1/4	2-1/2	2-1/2	2-13/16	3
Max. hole Ø (in)*	6-1/4	7-7/8	12-1/4	12-1/4	12-1/4	12-1/4	17-1/2	26
Max. jar load (lbf)	86,000	84,000	160,000	160,000	190,000	190,000	300,000	350,000
Tensile yield (lbf)	350,000	362,000	755,000	755,000	823,000	1,000,000	1,000,000	1,225,000
Torsional yield (lbf-ft)**	11,500	11,500	37,500	37,500	44,500	70,000	73,500	95,000
Total stroke (in)	17.5	19	22	22	20	20	21	21
Std up latch (lbf)	35,000	40,000	90,000	90,000	90,000	90,000	95,000	100,000
Max. up latch (lbf)	50,000	55,000	140,000	140,000	140,000	140,000	150,000	155,000
Std down latch (lbf)***	15,750	18,000	40,500	40,500	40,500	40,500	42,750	45,000
Max. down latch (lbf)	22,500	27,000	63,000	63,000	63,000	63,000	67,500	69,800
Pump open area (in²)	5.4	6.5	9.6	9.6	11.0	11.0	14.2	15.9
Length, latched (ft)	22	19.8	22.5	22.5	23.0	23.0	22.5	23.0
Weight (lb)	730	850	2,000	2,100	2,300	2,500	3,100	5,200
Ref.	4525	431	442	428	440	480	411	437

* Hole openers not recommended. ** Rating based on the yield of the body connections, independent of tool-joint connections. *** Down latch setting is 45% of the up latch setting, tolerance +/- 5.



DESIGN FEATURES

- **Delay in both directions:** hydraulic delay with mechanical latch, upstroke and downstroke.
- **Adjustable latch:** release settings can be changed up to the maximum values above.
- **Jet-type metering device** controls the hydraulic delay.
- **Safe on the rig floor:** no unexpected jarring while making connections or drilling.
- **Simple to operate:** no torque needed in the jarring procedure.
- **Built for high friction:** weight can be stacked before firing, even in horizontal sections.
- **Consistent firing:** continuous, repeatable jarring performance.

OPERATING PARAMETERS

Placement in string	Tension, neutral or compression
Delay	Hydraulic, up and down
Torque for jarring	None required
Tagging / picking up	No bleed-through needed
Down latch	45% of up latch, +/- 5
Hole openers	Not recommended

HOW TO ORDER

Please specify: 1. tool O.D. and ref. 2. top and bottom connections 3. up and down latch release settings (standard or adjusted, within the maximum) 4. hole diameter and drilling program (directional, horizontal or extended reach).